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How to Use
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About VectorBuilder

Codon Table

Codon tables are the Rosetta Stone of molecular biology; they are a fundamental tool in enabling you to decipher the genetic code. The table below acts as a reference for how nucleotides translate into codons as well as unique species-specific preferences for individual codons. Codon optimization can be used on large-scale sequences, while this table can be used to analyze individual codons. You can clone codon-optimized sequences easily into vectors and order downstream services by clicking [here](#).

Click [here](#) to design and order a vector with your chosen sequence in just a few clicks.
Receive plasmid preparations of your sequence at a variety of scales with fast turnaround.
Order high-titer packaging of your vector in ready-to-use virus.

Codon Table

Crash Course

Tips

Select organism: *

Human (Homo sapiens)

Table type: *

☒ Intuitive ☐ Proportional

1st Base	2nd Base (numbers indicate codon frequencies)				3rd Base
	T	C	A	G	
T	TTT Phe (F) 0.46	TCT Ser (S) 0.19	TAT Tyr (Y) 0.44	TGT Cys (C) 0.46	T
	TTC Phe (F) 0.54	TCC Ser (S) 0.22	TAC Tyr (Y) 0.56	TGC Cys (C) 0.54	C
	TTA Leu (L) 0.08	TCA Ser (S) 0.15	TAA Ter 0.30	TGA Ter 0.47	A
	TTG Leu (L) 0.13	TCG Ser (S) 0.05	TAG Ter 0.24	TGG Trp (W) 1.00	G
C	CTT Leu (L) 0.13	CCT Pro (P) 0.29	CAT His (H) 0.42	CGT Arg (R) 0.08	T
	CTC Leu (L) 0.20	CCC Pro (P) 0.32	CAC His (H) 0.58	CGC Arg (R) 0.18	C
	CTA Leu (L) 0.07	CCA Pro (P) 0.28	CAA Gln (Q) 0.27	CGA Arg (R) 0.11	A
	CTG Leu (L) 0.40	CCG Pro (P) 0.11	CAG Gln (Q) 0.73	CGG Arg (R) 0.20	G
A	ATT Ile (I) 0.36	ACT Thr (T) 0.25	AAT Asn (N) 0.47	AGT Ser (S) 0.15	T
	ATC Ile (I) 0.47	ACC Thr (T) 0.36	AAC Asn (N) 0.53	AGC Ser (S) 0.24	C
	ATA Ile (I) 0.17	ACA Thr (T) 0.28	AAA Lys (K) 0.43	AGA Arg (R) 0.21	A
	ATG Met (M) 1.00	ACG Thr (T) 0.11	AAG Lys (K) 0.57	AGG Arg (R) 0.21	G
G	GTT Val (V) 0.18	GCT Ala (A) 0.27	GAT Asp (D) 0.46	GGT Gly (G) 0.16	T
	GTC Val (V) 0.24	GCC Ala (A) 0.40	GAC Asp (D) 0.54	GGC Gly (G) 0.34	C
	GTA Val (V) 0.12	GCA Ala (A) 0.23	GAA Glu (E) 0.42	GGA Gly (G) 0.25	A
	GTG Val (V) 0.46	GCG Ala (A) 0.11	GAG Glu (E) 0.58	GGG Gly (G) 0.25	G



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